

Sampling Improvement in Software Engineering Surveys

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ABSTRACT

Context: Small and non-probabilistic samples represent relevant issues when discussing the external validity of empirical studies in Software Engineering. **Goal:** To investigate alternatives to improve the quality of samples (*size, heterogeneity and level of confidence*). **Method:** To replicate a survey on characteristics of agility in software processes by applying a systematic recruitment strategy over a professional social network. **Results:** It resulted in a sampling frame composed by 19 groups stratified according two perspectives: sharing of groups' members and main software engineering skills reported by the subjects. In total, 7,745 subjects were randomly recruited, resulting in 291 contributions. **Conclusions:** This sample was significantly larger, more heterogeneous and presents some strata with higher confidence levels than previous trials samples.

Categories and Subject Descriptors

D.2.4 [Software Engineering]: Software/ Program Verification—*statistical methods*.

General Terms

Experimentation, Human Factors.

Keywords

Experimental software engineering; sampling; population; survey; sampling frame.

1. INTRODUCTION

Small and non-probabilistic samples influence the external validity of empirical studies in Software Engineering (SE) including experiments [1,2] and surveys [3]. SE surveys are usually applied for gathering the opinion of researchers and practitioners regarding SE practices and technologies [3-6] or to support the collection of qualitative data in controlled experiments [7]. Therefore, we are investigating alternatives to improve the quality of samples (*size, heterogeneity and confidence level*) established for SE surveys. Into the context of our investigation, sample's *heterogeneity* is concerned with the geographical distribution of subjects, whereas *confidence level* is concerned with the experience regarding the survey context reported by subjects.

Our first experience toward sample's quality can be represented by

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a pilot study [5] using a recruitment plan for identifying a large set of members in the professional social network *LinkedIn* (www.linkedin.com). More than 870,000 members' profiles concerned with “software engineering” and distributed in different groups of interest represent *LinkedIn*'s SE community. The results allowed us to observe that samples collected from *LinkedIn* groups presented more heterogeneity and higher confidence than the ones obtained by convenience. As far as we are aware, it was the first reported experience concerning the application of sampling approaches based on social network populations over SE studies. This experience allowed us to identify opportunities to improve SE surveys by applying probabilistic sampling designs, including clustered and stratified sampling.

Continuing our line of investigation, we recently replicated a survey on characteristics of agility in software processes by performing a systematic recruitment of subjects through a relevant population retrieved from *LinkedIn*, presented in [8]. This survey was previously executed twice over populations established by convenience [6], with 230 invitations sent and only 25 participants in both trials. The results obtained by executing this third trial indicate both the feasibility of reducing the sampling bias and to significantly increase sample's size and heterogeneity, allowing its stratification.

Thus, this short paper presents our sampling experience in this replication, including a comparison between the samples from this third trial and previous ones. Section two briefly presents the study context and its hypotheses. Section three summarizes the recruitment plan, which has been fully described in a work fully devoted to it [8]. Section four presents the resulted samples, including a revision performed on the original stratification based on the obtained samples. Section five presents some lessons learned and section six presents observed threats to validity followed by the conclusions.

2. THE STUDY

In 2011, a web based survey was conducted to investigate the pertinence and relevance of agile practices and characteristics to introduce agility to software processes [6]. Two trials were performed with populations defined by convenience (P1) and predominantly composed by SE researchers. For our replication in 2013 (third trial), the population (P2) has been defined through a systematic recruitment plan, summarized in Section 3. Our interests, besides strengthening the survey's results, also include the observation of increasing sample's quality due the enlargement and generality of P2 when compared with P1. Thus, in order to compare the results extracted from both P1 and P2 samples (S1 and S2, respectively), the same hypotheses applied in our pilot study [5] were considered:

- H_01 : S1 and S2 present equivalent *heterogeneity*;

- H_{A1} : S2 is more *heterogeneous* than S1;
- H_{02} : S1 and S2 present equal *confidence level*;
- H_{A2} : The *confidence level* of S2 is higher than S1.

3. RECRUITMENT PLAN

A recruitment plan [8] was established to answer the following question: “Which are the LinkedIn groups concerned with Software Engineering and interested in agility characteristics and practices?”. To identify the available groups, it was organized a search string according the set of agile practices and characteristics presented in [6]. The application of this search string allowed us to identify 289 LinkedIn groups, from which 227 were excluded due to one or more exclusion criterion presented on Table 1.

Table 1. Incidence of each exclusion criterion.

Exclusion Criteria	#	% from Total
Local Groups	97	42.73%
Organizations, publicity and events	66	29.07%
Out of SE scope	33	14.54%
Vague description	25	11.01%
Single member groups	18	7.93%
Headhunting and job offering groups	8	3.52%
LinkedIn subgroups	3	1.32%
Non-English	1	0.44%
Total of Excluded Groups	227	78.55%

Considering the 62 remaining groups' members, 16 groups (3rd quartile) contained more than 90% of all retrieved gross population. So, to reduce our research efforts, the sampling frame was established based on these 16 groups. However, two of these groups were not available due to LinkedIn's subscription restrictions. Therefore, we decided to include the five highest groups from second quartile into the sampling frame, in order to preserve the minimal coverage of 90%. In November 2013, the 19 groups (sampling frame) totalized 264,540 LinkedIn members distributed over 149 countries from all continents. The LinkedIn filtering resources allowed the identification of 202,643 distinct members (net size), i.e., an overlapping rate of 30.54% ((gross size / net size) - 1). Thus, we re-grouped these 19 groups into clusters into clusters or strata based on their overlapping rates, i.e., to what percentage members of two groups are common [8]. Although it was not observed significant similarity by clustering the groups, it was observed the feasibility of regrouping them into eight strata (Table 2). We are aware that stratified sampling is commonly used to organize subjects under non-overlapped groups (such as citizens from distinct towns). However, we emphasize that overlapping of members among social network groups can happen and suggest the strength of similarity among such groups. A detailed description about this stratification can be found in [8].

4. STUDY RESULTS: Sampling Perspectives

The replication was executed between 5th December 2013 and 15th December 2013. In total, 7,745 subjects were randomly recruited from the sampling frame, taking into account the eight strata. It was collected 291 valid contributions during this period. Considering the high number of contributions (although one can notice the low participating rate), it was decided to review the stratification presented on Section 3 based in the main reported skills by 277 participants (95.19%), which answered the following open question: “What come into your mind when you think about your five main skills in software engineering?”. This question was inspired in the *free association method*, commonly applied into the context of social representation research to explore individual

cognitions [9,10]. Thus, after coding all 1,320 reported skills, it resulted in 325 distinct coded skills, distributed over 88 skill groups.

Table 2. LinkedIn Groups sampling frame and stratification.

Id	Stratum	Group Name
A	E1	Agile and Lean Software Development
B	E5	Test automation
C	E1	Scrum Practitioners
D	E1	Agile
E	E1	Agile Project Management Group
F	E5	Bug Free : Discussions in Software Testing
H	E7	Configuration and Release Management
I	E7	Configuration management
J	E3	Agile Testing
K	E8	Software Engineering Architecture Network
L	E1	Lean Agile Software Development Community
M	E1	Scrum Practitioners, Scrum Masters
N	E7	SCM Professional Network
P	E6	Software Engineers in Test
Q	E2	The Agile Project Management Hub
R	E1	Agilists
S	E3	eXtreme Programming (XP)
T	E4	Software Refactoring
U	E4	Test Driven Developers

Considering the context of this research, we decided to code into the skill group named “agile practices” every mention to *agile practices* [6], such as “refactoring”, “test driven development” and “product backlog”. In the same way, each social or personal skill presented in [6] as *characteristic of agility*, such as “being collaborative”, “leanness” and “adaptability” was included into the respective group named “characteristics of agility”. Explicit mentions to methods such as *Kanban*, *Scrum* or *XP* were coded into the “Agile Methods” skill group. Based on the sequence of skills answered by each subject (from the first to the fifth skill position) it was given a descendant grade from five points (first position) to one point (fifth position) for each skill. Then, it was observed that 90% of total points was attributed to the 20 most scored skill groups, which led us to evaluate the similarity between each stratum considering only these 20 skill groups. Table 3 presents the 10 most scored skill groups, including examples of skills for each group.

Table 3. Top 10 Skill Groups.

Skill Group	Coded Skills	Pts.	%
Personal Skill	Creativity, Learning, Planning	425	10.56%
Programming	Algorithms, Programming Languages	354	8.80%
SW Analysis and Design	OO Design, Design Patterns, SW Analysis	332	8.25%
Social Skill	Communication, Leadership	313	7.78%
SW Testing	Testing, Debugging	310	7.71%
Thinking and Reasoning	Abstraction, Analytical Thinking, Reasoning	251	6.24%
Agile Practices	Refactoring, TDD, Simple Design, Product Backlog	203	5.05%
Characteristic of Agility	Adaptability, Being Collaborative, Leanness	201	5.00%
SW Requirements	Requirements, Req. Analysis, Req. Elicitation	182	4.52%
SW Quality	Quality, Quality Assurance	147	3.65%

Thus, in order to evaluate the skill differences among the eight strata, it was depicted a matrix composed by the distribution of the 20 most scored skill groups in each stratum, as exemplified with the five most scored groups presented on Table 4. This matrix was submitted to a *hierarchical clustering analysis* by applying the

Pearson Correlation formula to calculate the distance between the 160 data points (combining 8 strata through 20 groups) and the *Average Linkage* algorithm to measure the similarity between clusters, calculated as the average of the distances between all the points in a pair of clusters [11]. Figure 1 shows the *dendrogram* generated by the *MiniTab* statistical tool, indicating the feasibility of grouping some pairs of strata having as threshold the same minimum similarity level (25%) applied to the original stratification [8].

Table 4. The 5 more scored skill groups over each stratum.

Stratum	Personal Skill	Progr.	SW An. and Design	Social Skill	SW Testing
E1	16.34%	7.48%	11.81%	22.31%	5.30%
E2	9.81%	10.62%	16.87%	16.73%	5.71%
E3	12.03%	8.46%	6.94%	20.49%	17.40%
E4	9.25%	20.56%	14.10%	14.81%	9.13%
E5	11.77%	2.83%	6.85%	14.80%	34.36%
E6	9.13%	22.01%	9.21%	3.88%	20.24%
E7	12.48%	14.36%	12.50%	0.00%	2.93%
E8	19.20%	13.67%	21.72%	6.98%	4.93%

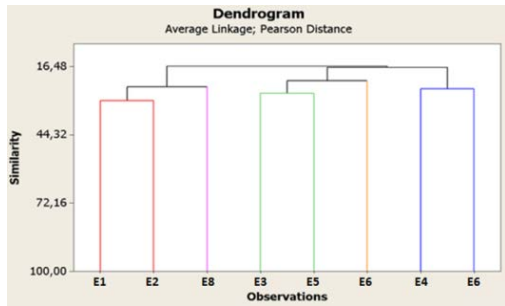


Figure 1. Clustering of the original strata.

Then, as a result, we decided to regroup the eight original strata into the following new five strata (St):

- St1 (E1+E2), composed by many *LinkedIn* groups concerned with agility in SE. Sample's main skill groups: *Personal Skills* (11.06%), *Social Skills* (10.38%) and *SW Analysis and Design* (9.10%).
- St2 (E3+E5), mainly composed by *LinkedIn* groups devoted to *Software Testing*, representing also the most relevant skill group (14.80%);
- St3 (E4+E6), mainly composed by *LinkedIn* groups devoted to *agile practices*, having *programming* as the most relevant skill group (15.76%);
- St4 (E7), composed by three *LinkedIn* groups concerned with configuration management (CM). Sample's main skills groups: *CM* (12.30%), *Programming* (10.73%) and *Personal Skills* (10.09%).
- St5 (E8), composed by a single *LinkedIn* group devoted to *software architecture*. Sample's main skill groups: *personal skills* (15.53%) and *SW analysis and design* (14.26%).

The following subsections will present the hypotheses testing considering S2 and the five new strata presented in this section.

4.1 Heterogeneity Analysis

The contributions regarding S2 were composed by 43 distinct countries, distributed in all continents (Table 5). Taking S1 into account, only nine countries can be identified in the reports of 12 participants. Notice that even if all S1 participants report a distinct country, it is not possible to overcome the heterogeneity of S2

regarding its countries' distribution. Thus, H_01 cannot be accepted and H_{A1} becomes valid. In practice, this result shows how the size of a sample and its origin can contribute to its heterogeneity.

Table 5. Distribution of S1 and S2 by Continent.

Continent	S1 (12 participants)	S2 (289 participants)
North America	41.7%	38.1%
Europe	33.3%	41.2%
Asia	25%	11.8%
Latin America	-	5.9%
Oceania	16.7%	2.1%
Africa	-	0.1%

Additionally, it was observed that the distribution of members from the 10 most represented countries in P2 (USA, India, UK, Canada, Australia, Brazil, Netherlands, Sweden, France and China) presents high correlation (95.1%, p-value <0.001) with the distribution of participants from the same countries in S2 (Pearson's correlation test). This result suggests the participation in the survey was not biased due to language barriers, since 5 of these 10 countries are not English native. Although one can also evidence the samples heterogeneity through the presented stratification it is not possible to compare these results with S1 since the main S1 participants' SE skills were not collected.

4.2 Confidence Analysis

The confidence level of S1 and S2 was analyzed through the distribution of experience level, calculated for each subject using the values given by each one with the following formula (1):

$$Exp = (L(Acad) + L(AgPap) + L(AgExp) + S(AgProj)) / Max \quad (1),$$

where: *Acad*= Academic Degree, *Likert* scale (0-Undergraduation, 1-Specialization, 2-Master Degree, 3-DSc/Phd Degree); *AgPap*= number of publications related with agile context, scale (0-3); *AgExp*=Agile experience, *Likert* Scale (0-Low, 1-Medium, 2-High, 3-Very High); *AgProj*= Amount of Agile Projects, scale (0-3) and *Max*= numerator max value (12). Both *AgProj* and *AgPap* use the same scale: (0-none; 1-first quartile; 2-second quartile; 3-third quartile). This formula represents an evolution of the original one [6]. The change was necessary because the original formula can introduce an overestimating bias due the lack of scalability in the number of projects reported by each subject.

After calculating the confidence levels for S1 and S2, removing outliers and verifying both S1 and S2 distributions were normal (*Kolmogorov-Smirnov*) and homoscedastic (*Levene*), it was applied the Student-t test to evaluate whether S2 presents a confidence level significantly higher than S1. No significant difference between S1 and S2 was observed (Table 6- The number of removed outliers is indicated in parenthesis besides each sample size). Therefore, we can't reject H_02 .

Table 6. Comparison of Experience Level distributions.

Sample	#	Mean	Normal (KS)	Student t-Test	Mann-Whitney test
S1	25(1)	0.375	Yes	-	-
S2	291	0.418	Yes	0.055	-
S2-St1	97	0.460	Yes	0.011(0)	-
S2-St2	81(3)	0.382	Yes	0.424(3)	-
S2-St3	57(7)	0.470	Yes	0.003(7)	-
S2-St4	24	0.333	No	-	0.415 (0)
S2-St5	31	0.403	No	-	0.171 (0)

In addition, it was also compared the distribution of experience levels between S1 and each S2's stratum. As a result, it was

observed (Student-t test) that S2-St1 and S2-St3 present distributions of samples significantly more experienced than S1. Figure 2 shows both boxplots for these distributions. Thus, these results suggest a correspondence between the characteristics of each stratum and the calculated experience levels.

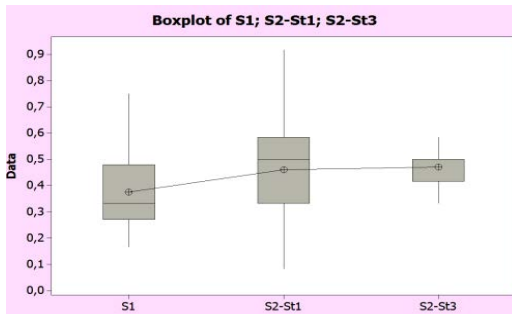


Figure 2. Experience Level distributions for S1,S2-St1, S2-St3.

5. LESSONS LEARNED

The analysis performed in this study allowed us to observe and learn some lessons regarding the planning of populations in SE surveys, as following:

- Although members' overlapping can suggest similarity between groups, it is risky to only depend on this single feature to perform the stratification;
- It is worthwhile to deep investigate available groups on professional social networks, adding new variables for supporting the stratification. As an alternative to get more information about each group of subjects, we suggest first to invite the subjects to answer a brief characterization questionnaire and next to recruit them to the survey, as usually performed in SE experiments;
- Since social network groups are naturally (by participant's interests) established, we suggest to initially consider each group as a distinct stratum and next to evaluate their similarities. Thus, for subjects' characterization, we suggest calculating each sample's size based in the group size. In the worst case, each group can be maintained as a single stratum;
- A simple and optional open question can make a significant difference and contribute to trace subjects' profile without overloading them with tough questionnaires, especially when there are some data naturally available and updated in the source of recruitment. However, the coding process in large scale can be exhaustive and it must be considered in the study planning;
- Regarding the sample's size, we observed that spontaneous collaboration on random recruitments tends to be smaller than when made by convenience. In our three last experiences with recruitments through professional social networks (*LinkedIn* and *ResearchGate*), we observed a participation rate between 3% and 4% from all invitations sent. However, a small percentage in the large can be better than a large percentage in the small.

6. THREATS TO VALIDITY

We emphasize the *LinkedIn's* limitation on restricting the retrieving of all members' data as a threat to validity, which led us to consider each group (not each member) as a single unit. Additionally, one can see that the concepts of "heterogeneity" and "confidence" was based on small subsets of variables which values were given by own subjects. The coding of SE skills can also be considered a threat, since it was based in the interpretation

of the researchers over each expression reported by the subjects. The technical limitations for performing large-scale recruitment directly by using *LinkedIn* resources also represents a threat, since it was needed to use macros and intermediary databases, which introduced a set of operational risks. Finally, we recognize the low number of analyzed variables in order to support the stratification and also the establishment of new strata after sampling.

7. CONCLUSIONS

We are far from any default solution to enlarge the number of subjects and increase sampling quality in SE studies based in the probabilistic context. One can see that the contribution rate of social networks members was very small and far from our "natural, and maybe naïve," idea of getting "large scale" samples by using such technologies. However, the results from this study suggest that we can evolve the current scenario of typical samples by convenience in SE surveys through applying systematic recruitment strategies over alternative sources of recruitment. As a next step, we intend to develop and evaluate a conceptual framework to support SE researchers in the investigation and establishment of adequate samples for their surveys.

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